

Chapter 3

Lecture 4

Data Modeling Using the Entity-Relationship (ER) Model

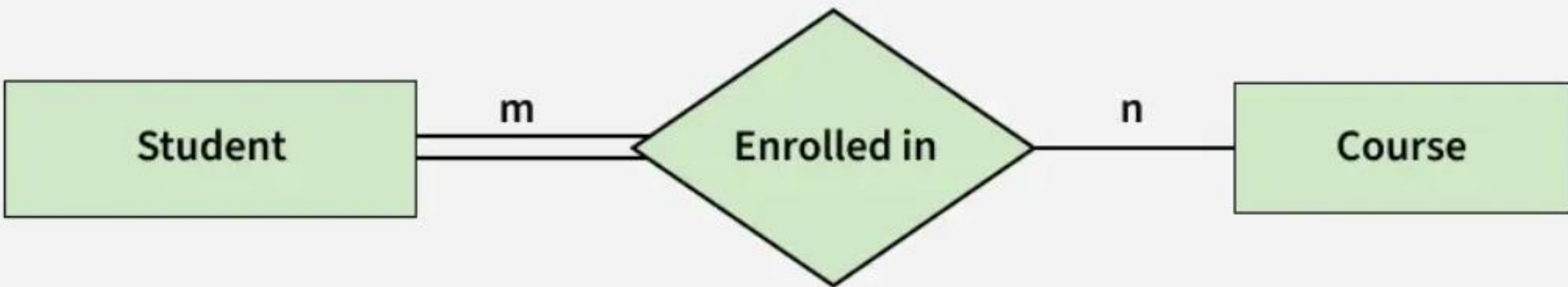


5th Edition

Elmasri / Navathe

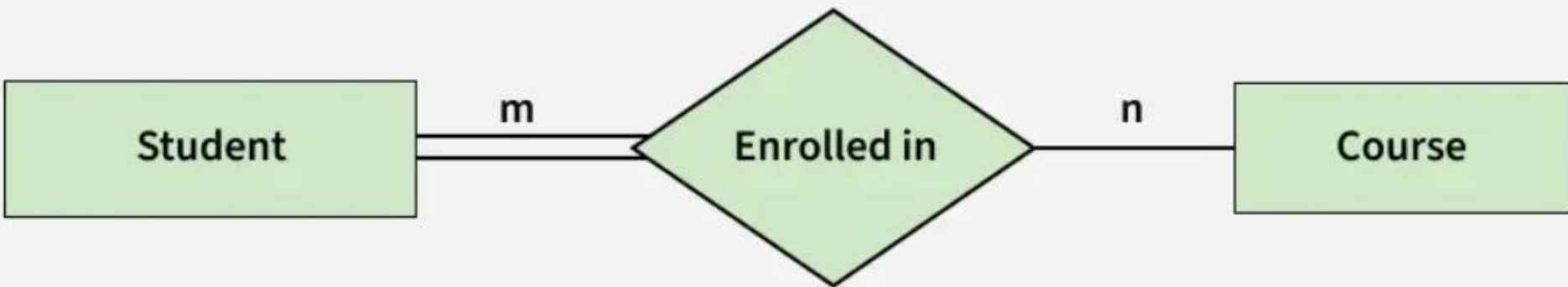
Participation Constraint

- Total Participation: Each entity in the entity set must participate in the relationship. If each student must enroll in a course, the participation of students will be total. Total participation is shown by a double line in the ER diagram.



Participation Constraint

- Partial Participation: The entity in the entity set may or may NOT participate in the relationship. If some courses are not enrolled by any of the students, the participation in the course will be partial.



Example COMPANY Database

- We need to create a database schema design based on the following (simplified) **requirements** of the COMPANY Database:
 - The company is organized into DEPARTMENTS. Each department has a name, number and an employee who *manages* the department. We keep track of the start date of the department manager. A department may have several locations.
 - Each department *controls* a number of PROJECTs. Each project has a unique name, unique number and is located at a single location.

Example COMPANY Database (Contd.)

- We store each EMPLOYEE's social security number, address, salary, and birthdate.
 - Each employee *works for* One department but may *work on* several projects.
 - We keep track of the *number of hours per week* that an employee currently works on each project.
 - We also keep track of the *direct supervisor* of each employee.
- Each employee may *have* a number of DEPENDENTS.
 - For each dependent, we keep track of their name, sex, birthdate, and relationship to the employee.

Initial Design of Entity Types for the COMPANY Database Schema

- Based on the requirements, we can identify four initial entity types in the COMPANY database:
 - DEPARTMENT
 - PROJECT
 - EMPLOYEE
 - DEPENDENT
- Their initial design is shown on the following slide
- The initial attributes shown are derived from the requirements description

Initial Design of Entity Types:

EMPLOYEE, DEPARTMENT, PROJECT, DEPENDENT

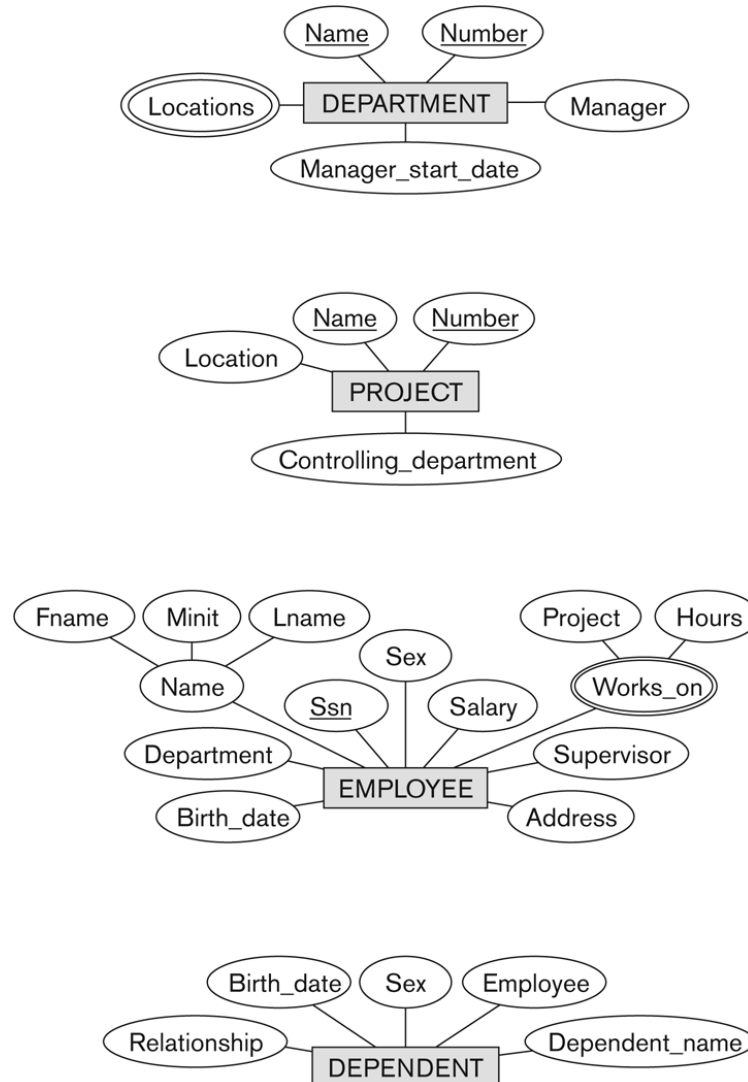


Figure 3.8

Preliminary design of entity types for the COMPANY database. Some of the shown attributes will be refined into relationships.

ER DIAGRAM – Relationship Types are:

WORKS_FOR, MANAGES, WORKS_ON, CONTROLS, SUPERVISION, DEPENDENTS_OF

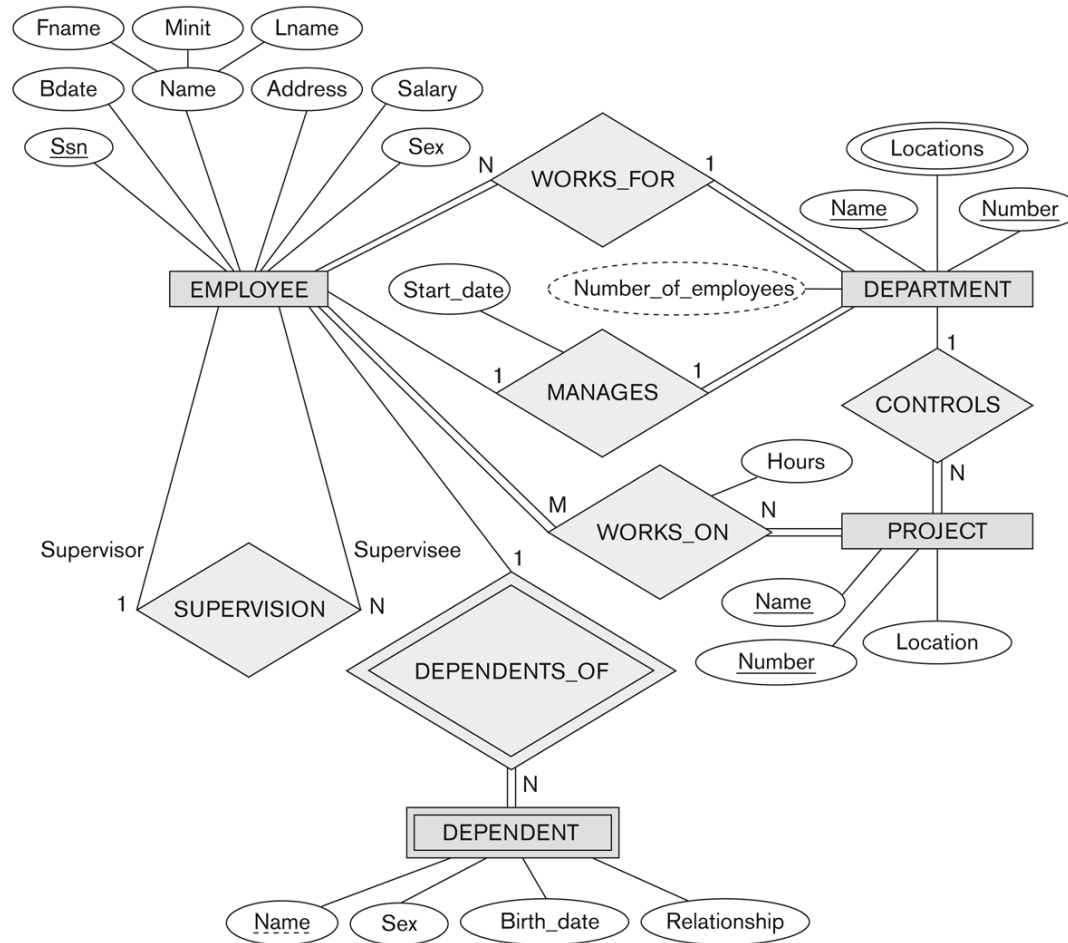


Figure 3.2

An ER schema diagram for the COMPANY database. The diagrammatic notation is introduced gradually throughout this chapter.

2. Relationships (Diamonds)

WORKS_FOR

- Between EMPLOYEE (N) and DEPARTMENT (1)
- Meaning:
 - Each employee works for **one** department
 - A department has **many** employees

Participation by Relationship

WORKS_FOR (EMPLOYEE — DEPARTMENT)

- EMPLOYEE: Total participation
 - Every employee must work for a department
- DEPARTMENT: Partial participation
 - A department may exist with no employees

◆ Conceptual (real-world) reason

In this company database:

- Every employee **must** be assigned to a department
- There is **no such thing** as an employee without a department

Because this rule is **mandatory**, the figure shows **total participation**.

Discussion on Relationship Types

MANAGES

- Between **EMPLOYEE (1)** and **DEPARTMENT (1)**
- Attribute:
 - **Start_date** the date on which an **EMPLOYEE** started managing a **DEPARTMEN**
- Meaning:
 - Each department is managed by **one** employee
 - An employee manages **at most one** department

MANAGES (EMPLOYEE — DEPARTMENT)

- **DEPARTMENT: Total participation**
 - Every department must have a manager
- **EMPLOYEE: Partial participation**
 - Not every employee is a manager

Discussion on Relationship Types

CONTROLS

- Between **DEPARTMENT** (1) and **PROJECT** (N)
- Meaning:
 - One department controls many projects
 - Each project is controlled by exactly one department

CONTROLS (DEPARTMENT — PROJECT)

- **PROJECT: Total participation**
 - Every project must be controlled by a department
- **DEPARTMENT: Partial participation**
 - A department may control zero projects

WORKS_ON

- Between **EMPLOYEE (M)** and **PROJECT (N)** (many-to-many)
- Attribute:
 - Hours (how many hours the employee works on that project)

WORKS_ON (EMPLOYEE — PROJECT)

- **EMPLOYEE: Partial participation**
 - An employee may work on no projects
- **PROJECT: Partial participation**
 - A project may have no employees assigned

When *would* WORKS_ON be total?

It would be **total** if the rules were:

- Every employee must work on **at least one project**
- Every project must have **at least one employee**

In that case, the diagram would show **double lines** on both sides.

Discussion on Relationship Types

WORKS_ON

- Between **EMPLOYEE (M)** and **PROJECT (N)** (many-to-many)
- Attribute:
 - Hours (how many hours the employee works on that project)

WORKS_ON (EMPLOYEE — PROJECT)

- **EMPLOYEE: Partial participation**
 - An employee may work on no projects
- **PROJECT: Partial participation**
 - A project may have no employees assigned

Discussion on Relationship Types

SUPERVISION (recursive relationship)

- EMPLOYEE ↔ EMPLOYEE
- Roles:
 - Supervisor (1)
 - Supervisee (N)
- Meaning:
 - One employee can supervise many employees
 - Each employee has at most one supervisor

SUPERVISION (EMPLOYEE — EMPLOYEE)

- Supervisor role: Partial participation
→ Not every employee supervises others
- Supervisee role: Partial participation
→ Top-level employees may have no supervisor

UML class diagrams

- Represent classes (similar to entity types) as large rounded boxes with three sections:
 - Top section includes entity type (class) name
 - Second section includes attributes
 - Third section includes class operations (operations are not in basic ER model)
- Relationships (called associations) represented as lines connecting the classes
 - Other UML terminology also differs from ER terminology
- Used in database design and object-oriented software design
- UML has many other types of diagrams for software design (see Chapter 12)

UML class diagram for COMPANY database schema

Figure 3.16

The COMPANY conceptual schema in UML class diagram notation.

